

Concept of Operations — Appendix

ESPERANZA II · NASA Mars to Table Challenge 2026 · ESPERANZA Research Team · María Jesús Puerta Angulo
Deliverable 3.2 · Issued 19 July 2026 · Supports Deliverable 3.1 (Concept of Operations slide deck)

Scope. This appendix substantiates figures presented in the Concept of Operations deck. It introduces no new concepts. Each section states the **basis of estimate** for its values so that a reviewer can trace and verify them, and every figure is reconciled against its neighbours: lighting, water, heat, CO₂ and crew time are treated as one system, not as independent budgets.

1. Electrical power — input stream

Food-system demand is **58.1 kW mean, 64.0 kW peak, 77 kW installed design capacity** (peak × 1.20 margin). Appendix B of the Rules states power is **not constrained**; the team therefore estimates and justifies demand rather than designing to a cap, and proposes surface fission power of matching class.

Line item	kW	Basis of estimate
Six modules — LunaRex 0.6 · MarsRem 2.1 · HybridSoil 0.4 · CyanoGrow 4.8 · FermentLab 1.9 · NEXUS 0.9	10.7	Solution Summary §4 module table
Canopy LED	30.4	$\Sigma(\text{DLI} \times \text{area})$ over 373 m ² ÷ 3.0 μmol/J ÷ 88,775 s
Canopy HVAC / ventilation	5.0	Fans only, 8% of lighting load, × 1.84 hypobaric correction (p 0.667 vs 1.225 kg/m ³)
Dehumidification / condensate	2.2	17.8 kW latent rejected through a passive loop to the Martian sink; pumping power only
Thermal loop and pumps	2.5	≈4% of rejected thermal load, pumped single-phase parasitics
Nutrient and aeration pumps, 373 m ²	2.5	5–8 W/m ² , hydroponic practice
Mushroom substrate pasteurisation	1.2	Batch process, sol-averaged
Work and corridor lighting	0.8	Task and egress illumination, growing volume
Galley — mill, griddle, centrifuge	0.7	Equipment inventory, Solution Summary power box
Mushroom chamber environment	0.6	90% RH, elevated air-change rate
Refrigerated store, 4 °C	0.8	Storage condition required by Deliverable 2.2
Water treatment and condensate UV	0.4	Potable polishing of recovered condensate
Martian CO ₂ capture and compression	0.3	3.1 kg/sol, 0.6 → 56.5 kPa (94:1). Isothermal ideal work 8.8 W; ~7× margin retained for adsorption, valves and control
Total, mean	58.1	Peak 64.0 · installed 77

No double counting. Spirulina photobioreactor illumination, mixing and sparging are internal to CyanoGrow (4.8 kW) and are **not** part of the 30.4 kW canopy figure — the canopy is the 373 m² of upper-tier crops only. *Tenebrio* rearing is internal to FermentLab (1.9 kW). Galley load is inside the 58.1 kW total.

1.1 Peak flattening is a procedure, not a coincidence

Zone	m ²	DLI	Photoperiod	Instantaneous kW	Mean kW
Wheat, USU-Apogee	150	24	24/0 continuous	13.52	13.52
Soybean	80	22	12.33 / 12.33	13.22	6.61
Potato	80	20	12.33 / 12.33, opposite window	12.02	6.01
Tomato, pepper, leafy greens	63	14–22	16/8	6.55	4.25
Canopy	373			33.3 peak	30.4 mean

Wheat runs continuously as constant baseload — USU-Apogee was bred for controlled environments and tolerates continuous light without the apical chlorosis seen in other cultivars. Soybean and potato tessellate opposite half-sol windows. Peak therefore sits only **9% above mean**, which is what allows a modest plant to serve a 373 m² farm.

Photoperiod is not a free parameter. Soybean is a short-day species: under a 16 h photoperiod it does not flower, which removes tofu, tempeh and miso — 345 g per crewmember per sol of the menu. Potato tuberisation likewise favours short days. The photoperiod schedule is therefore a crop-physiology requirement first and a load-management benefit second.

1.2 Lighting architecture — why conducted daylight was traded out

An earlier iteration carried conducted Martian sunlight supplying part of the canopy Daily Light Integral. It was withdrawn on three grounds, in order of weight:

- **Mission reliability.** Global dust storms suppress surface PAR by more than 90% for weeks at a time, without reliable forecast. A conducted-light system must therefore carry the full LED capacity as backup regardless — the average-case saving does not remove the installed cost.
- **Pressure-shell integrity.** Optical conduction requires penetrations of the pressure shell and of the radiation shielding, adding leak paths and MMOD exposure to the most safety-critical boundary of the habitat.
- **Thermal balance.** A translucent boundary cannot be insulated and transmissive at once. Over 373 m² at $\Delta T \approx 82$ K the conductive loss is 15–46 kW depending on glazing ($U = 0.5\text{--}1.5$ W/m²K) against roughly 17 kW of optical gain. Under Martian conditions the range is wide, but it does not favour the transparent envelope.

LED sole-source decouples crop Daily Light Integral from Mars weather at a fixed, schedulable electrical load.

2. Water — input and output streams

Stream	Rate	Basis of estimate
Canopy transpiration, gross	644 L/sol	30.4 kW $\times$ 0.90 absorbed $\times$ 0.65 to latent $\div$ 2,450 kJ/kg. In a closed chamber the evaporation energy can only come from absorbed light
Condensate recovered	634 L/sol	98.5% recovery
Net make-up required	9.7 L/sol	644 $\times$ 1.5%
Grey water received from ECLSS	12 L/sol	Habitat interface, Solution Summary §4
Nutrient solution standing volume	12,000 L	32 L/m ² sealed inventory. This is an inventory, not a daily flow

Correction of record. Earlier versions of this appendix stated **3,226 L/sol** of transpiration. Evaporating that mass requires 89 kW of latent heat while the lighting delivers 30.4 kW — thermodynamically impossible in a closed chamber. The 3,226 figure corresponds to a canopy at DLI $\approx$ 120–130; this farm operates at DLI 14–24. All references to 3,226 L/sol and to a 50 L/sol ISRU water input are withdrawn. The revised figure closes against the lighting budget, which is how the error was found.

2.1 Condenser anti-icing

Rejecting latent heat against a loop near Martian ambient produces ice rather than liquid water. Condenser control is by hot-gas bypass and mixing valve with a setpoint of 4–10 °C, and a trip if loop temperature falls below 2 °C. Recovered condensate is UV-polished before return to potable service.

3. Atmosphere — the balance closes, and it closes the other way

Stream	Rate	Basis of estimate
CO ₂ fixed by crops	13.5 kg/sol	307 mol C → 8.77 kg dry biomass/sol = 23.5 g/m ² /sol over 373 m ²
CO ₂ fixed by spirulina (40 m ² PBR)	7.2 kg/sol	35% of total fixation; omitted from earlier revisions of this table
Total CO₂ fixed	20.7 kg/sol	13.5 crops + 7.2 spirulina
CO ₂ produced by crew of 15	15.4 kg/sol	Metabolic, NASA-STD-3001 baseline values
Net carbon balance	+0.6 kg/sol surplus	21.3 produced (15.4 crew + 1.5 fermentation + 0.8 composting + 0.5 heterotrophs + 3.1 Mars) vs 20.7 fixed
O ₂ released by crops	9.82 kg/sol	Photosynthetic stoichiometry from 13.5 kg CO ₂
O ₂ released by spirulina (40 m ² PBR)	5.24 kg/sol	Photosynthetic stoichiometry from 7.2 kg CO ₂ (×32/44)
O ₂ consumed by crew	12.95 kg/sol	Metabolic
Total O₂ released	15.02 kg/sol	116% of crew demand — crops alone 76% (9.82 / 12.95)
Martian CO ₂ make-up	3.1 kg/sol	See note below

On the CO₂ make-up. Crew respiration (15.4 kg/sol) is the primary carbon feed, supplying **74%** of the 20.7 kg/sol that crops (13.5) and spirulina (7.2) fix. Fermentation, composting and heterotrophs return a further 2.8 kg/sol. The remaining **2.5 kg/sol** is drawn from the Martian atmosphere, which is 95% CO₂ — 3.1 kg/sol extracted, 0.6 buffered. **Using in-situ CO₂ to feed photosynthesis is a design choice, not a life-support shortfall:** the crew is not a large enough carbon source to saturate 413 m² of photosynthetic area, and on Mars the cheapest way to close that gap is the atmosphere outside the door. The full ledger is in ConOps Appendix A3.

3.1 Crop chambers hold an atmosphere independent of the habitat

Crop chambers operate at **56.5 kPa, ~21% O₂, 1,200–1,500 ppm CO₂**, airlock-separated from the habitat and exporting O₂ to it.

Wheat, soybean, potato, tomato and pepper are all C3 species. At the habitat's 34% O₂ the oxygenation reaction of Rubisco competes more strongly with carboxylation, and net photosynthesis falls on the order of 15–25% — which would make every yield in the Meal Plan optimistic. Holding the chambers near 21% O₂ with elevated CO₂ removes that penalty. The same separation resolves a second issue: LED driver electronics are housed in a normoxic bay with DC distribution to the luminaires, avoiding flammability concerns in a hyperoxic volume per NASA-STD-6001.

4. Heat rejection

Essentially all electrical input becomes heat, less the fraction stored as chemical energy in biomass: 58.1 kW electrical – 1.7 kW stored = **56.4 kW thermal to reject**.

Parameter	Value	Basis of estimate
Radiator area	~300 m ²	Sized for the dusty case, not the clean case
Loop temperature	40 °C	Raising rejection temperature reduces area as T ⁴
Orientation	Vertical panels, N–S	Minimises solar view factor
Surface	Low absorptivity, α ≈ 0.15 clean	Martian dust deposition drives α toward ≈0.55

Why 300 m² and not 220. A clean radiator at mean conditions would need roughly 220 m². Under dust deposition the absorbed solar term rises until a 230 m² array fails to reject at local noon. The array is therefore sized for the degraded case and paired with a cleaning protocol. **Those cleaning EVA hours are charged back to the calorie budget at 200 kcal per EVA hour,** closing the loop between lighting, heat, maintenance and nutrition.

5. Crew time — the binding operational constraint

Appendix B of the Rules constrains mass, power, volume and water with the phrase *not constrained*. It constrains exactly one resource with *must*: *"All components of continuously maintaining the food system must be executable within a typical workday/workweek for the food systems engineer and nutrition/meal prep specialist... 45-hour workweek over five Martian sols... should not require overtime."*

With two specialists the budget is **90 h per rolling five-sol window**. Compliance is assessed on that window, not on any single sol, and it is assessed on the **specialists' own load** — not on the whole-crew galley total, part of which is carried by the all-crew cooking rotation.

Metric	Value	Basis of estimate
Galley load, mean	15.40 h/sol	Measured from Deliverable 2.2 under the batch protocol below
Peak sol	19.03 h (Sol 3)	Second peak 18.72 h (Sol 13)
Minimum sol	13.35 h (Sol 6)	
Worst rolling five-sol window, galley, all crew	80.3 h	Whole-crew galley total; not the compliance metric
— of which carried by the all-crew cooking rotation	48.9 h/cycle	Earth-ingredient prep 4.7 h + cooking 31.2 h + galley cleaning 13.1 h $\approx$ 45 min/person/sol across the other 13 crew
— of which retained by the two specialists	28.1 h/cycle	In-situ ingredient processing: fermentation, spirulina harvest, <i>Tenebrio</i> and mushroom handling, soy and grain processing
Specialist non-galley tasks	49.0 h/cycle	Crop tending and harvest 22.0 · HACCP, water and perchlorate monitoring 9.0 · maintenance 7.0 · documentation 6.0 · planning and inventory 5.0
Specialist load, worst rolling five-sol window	81.2 h of 90 h — 90%	Mean 77.1 h of 90 (86%); worst single specialist 40.6 h of 45; 0 of 10 windows exceed budget; no overtime required

5.1 Batch-processing accounting

Times in Deliverable 2.2 are charged per ingredient-*event* under a mise en place galley protocol, not per ingredient-*line*:

- **Preparation** — once per ingredient per sol. All portions of a given ingredient required across that sol's services are washed, weighed and cut in a single session at the start of the sol.
- **Cooking** — once per ingredient per meal. Cooking occurs at each service and is not shared across services.
- **Cleaning** — once per ingredient per sol, matching the preparation batching.

The unbatched line-by-line sum totals 21.03 h/sol, which double-charges mise en place for ingredients recurring across a sol's services — soybean oil alone appears on 130 ingredient-lines across the cycle.

This is the design driver of the NEXUS layer. With mass, power and volume unconstrained, crew time is the resource that cannot be expanded. NEXUS exists to buy it back: it lets two specialists manage a system that would otherwise require six to eight operators. Growing more of a crop is free in power and volume and expensive in hours.

6. Earth-provisioned energy audit

The Rules cap Earth-provisioned food at 50% **by calories**. ESPERANZA II reports **47.0% by calories** on the conservative crop-table nameplate basis — the figure carried in the Solution Summary and the ConOps. Measured against the audited 14-sol menu the same system is **41.2%**; both bases are set out and reconciled below, and the headline uses the higher of the two.

6.1 Method

Deliverable 2.2 records an ingredient source and a mass for all 1,004 ingredient lines but carries no energy column; Deliverable 2.1 records calories per meal but no source attribution. The ratio the Rules specify is therefore not directly readable from either sheet. Each of the 177 unique ingredients was assigned an energy density from USDA FoodData Central reference items, volumetric entries were converted to mass (only cooking oil using a non-unity density, 0.92 g/mL),

and line energy was aggregated by source. The per-ingredient table with reference item and confidence flag accompanies the Meal Plan.

6.2 Validation

The method reconstructs the plan's own nutrition without being fitted to it: **2,992 kcal per crewmember per sol against the 3,035 kcal target — 98.6% agreement**. This validates both the energy table and the interpretation of Deliverable 2.2 quantities as per crewmember per sol.

Source	kcal per crewmember, 14 sols	Share
Grown in-situ	24,622	58.8%
Shipped from Earth, bulk	17,263	41.2%
Total	41,885	100%

Per-sol range is 32.9% to 47.2%: **no individual sol exceeds the cap**, so compliance holds not only on the 14-sol average but on every sol of the cycle. For reference only, the same audit by mass yields **16.1%** — computed on the same basis as the energy audit, with volumetric entries converted at their stated density (cooking oil at 0.92 g/mL); an earlier revision reported 16.4% because it summed the raw g/mL column without that conversion. Mass is not the metric the Rules specify and is stated solely to prevent the two figures being confused.

6.3 Declared limitations

- Of 177 unique ingredients, 62 carry high confidence, 57 medium and 58 low. Weighted by contribution, 67% of calories come from high-confidence entries and 6.4% from low-confidence entries, the latter dominated by spice blends of 1–3 g.
- The largest single uncertainty is *Tenebrio molitor* flour at 5.20 kcal/g, which has no direct USDA entry; the value derives from the declared composition of approximately 50% protein and 30% fat.
- **Two bases, both declared.** The 41.2% below is measured against the *audited 14-sol menu*. The Solution Summary and the ConOps report **47.0%**, measured against the *crop-table nameplate* of 24,141 kcal/sol: the 2,628 kcal/sol by which the menu draws above that nameplate is booked to Earth-provisioned stock as a declared reserve rather than claimed as production. Both figures are correct on their own basis, and the more conservative one (47.0%) is the one carried in the headline. Cooking oil is 3% of ingredient mass but 15% of plan energy and is Earth-provisioned. It is therefore the dominant lever on the 41.2% figure and the primary target for in-situ pressing in a future iteration.

7. Area and volume

Zone	m ²	Note
Photosynthetic canopy, 11 crops	373	The only area drawing canopy lighting
Spirulina photobioreactor	40	Illumination internal to CyanoGrow
<i>Tenebrio</i> rearing racks	48	Heterotrophic — no photosynthetic light
Mushroom racks	50	Heterotrophic — no photosynthetic light
Total growing area	511	
Crew Food Deck — galley 14 · dining 22 · dry pantry 8 · cold store 6 · seed and culture bank 4	54	Preparation and consumption volume per Rules Appendix B

Earlier documents stated 481 m², which reconciled with no breakdown. The figure is corrected to 511 m² with the components shown so that a reviewer can sum them.

8. Technology maturity and gaps

Element	TRL	Maturity basis	Gap to close
Regolith mineral mapping (LunaRex)	3	Random-forest and Gaussian-process classification of regolith composition from published lunar and Martian datasets, 77% accuracy. Analytical proof of concept only	No beneficiation or sorting hardware built or tested; the classifier is trained on published datasets rather than on in-situ assay. A breadboard demonstration on simulant is required before TRL 4 can be claimed
LED sole-source crop production	6–7	Flight heritage in Veggie and the Advanced Plant Habitat aboard ISS; commercial vertical farming at scale	Fixture lifetime qualification to ≥50,000 h under Mars surface conditions
Hydroponic staple crops (wheat, soybean, potato)	5–6	Extensive controlled-environment literature; potato under NASA-funded CELSS work	Yield confirmation at 56.5 kPa hypobaric pressure
Cyanobacterial photobioreactor	4–5	Terrestrial <i>Arthrospira</i> production is industrial; N ₂ fixation by <i>Anabaena</i> demonstrated in the laboratory	Long-duration culture stability and strain drift over multi-year operation
Perchlorate bioremediation (MarsRem)	4	Perchlorate-reducing metabolism characterised in <i>Azospira suillum</i> ; pcrA/cld pathway published	Rate and completeness in Martian regolith matrix; process scale-up
Regolith-derived growth substrate (HybridSoil)	4	Simulant studies for both lunar and Martian regolith	Validation on returned or in-situ material rather than simulant
<i>Tenebrio molitor</i> rearing	5–6	Established commercial industry; feed-conversion data well characterised	Closed-loop rearing on in-situ inedible biomass; microgravity-independent but Mars-gravity behaviour unverified
Fermentation (miso, tempeh, kefir, bread)	6–7	Millennia of terrestrial practice; process control well understood	Starter culture viability across multi-year storage and refresh cadence
NEXUS control layer	3–4	Component algorithms established; sensing hardware flight-proven. Inference step demonstrated by the author on the SpaceX Inspiration4 public archive (NASA OSDR, OSD-569 to OSD-687, n = 4): multi-modal fusion separates pre- from post-flight state at LOSO accuracy 0.708 against a permuted-label null of 0.489 ± 0.107 , $p = 0.026$ (Puerta Angulo, 2026, preprint, DOI 10.21203/rs.3.rs-10232328/v1)	Integrated closed-loop validation with human subjects in an analogue environment. The cited work establishes the inference step on a spaceflight analogue but does not close the loop to meal actuation, and is a preprint under review
Surface fission power	5–6	KRUSTY demonstration and the ongoing Fission Surface Power project	Not in scope for this challenge; cited as heritage for the class of plant assumed

Overall system TRL is governed by its least mature elements — LunaRex at TRL 3, NEXUS at TRL 3–4, and MarsRem and HybridSoil at TRL 4. The remaining architecture rests on technologies at TRL 5 or above. The gaps above are stated so that a reviewer can judge the development path rather than infer it.

9. Anticipated reviewer questions

Your crew produces 15.4 kg of CO₂ per sol but your system fixes 20.7. Where does the missing carbon come from, and does that mean the loop is not closed?

From the Martian atmosphere, and no. The crew supplies 74% of the carbon; fermentation, composting and heterotrophs return 2.8 kg/sol more; the remaining 2.5 kg/sol is extracted from an atmosphere that is 95% CO₂ at 3.1 kg/sol, with 0.6 buffered. Fifteen people are simply not a large enough carbon source to saturate 413 m² of photosynthetic area — and that is the point of siting the farm on Mars rather than shipping the carbon. The loop is closed on the resources that are scarce (water 98.5%, organics 98.3%); it is deliberately open on the one resource that is abundant outside the door. Earlier revisions of this table omitted the spirulina photobioreactor, which fixes 7.2 kg/sol — 35% of the total — and therefore reported a surplus where the full balance shows a fixation deficit met in situ. Section 3 and ConOps Appendix A3 state the corrected ledger.

Your crops are all C3 and your habitat is at 34% O₂. Have you modelled photorespiration? If the chambers share the habitat atmosphere, your yields are overstated.

They do not share it. Crop chambers hold an independent atmosphere at approximately 21% O₂ with CO₂ elevated to 1,200–1,500 ppm, airlock-separated and exporting O₂ to the habitat. Section 3.1 gives the rationale and the magnitude of the penalty avoided.

Your radiator assumes a clean surface. Give us the area at local noon with Martian dust deposition, when absorptivity rises from 0.15 to about 0.55.

That is why the array is 300 m² and not the 220 m² a clean-case calculation yields, and why the loop runs at 40 °C with vertical N–S panels. A 230 m² array does fail to reject at local noon under deposition. Cleaning is a scheduled EVA task and its hours are charged back to the calorie budget at 200 kcal per EVA hour. Section 4 gives the sizing.

How many crew hours per sol does it take to operate six modules with fifteen people?

Galley operations are 15.40 h/sol measured across all crew, of which the two specialists retain the in-situ processing share (28.1 h per five-sol cycle) and the all-crew cooking rotation carries the routine remainder (48.9 h, ≈45 min/person/sol). With 49.0 h/cycle of non-galley specialist tasks on top, the specialists run at 77.1 h of the 90 h Appendix B budget (86%), worst window 81.2 h (90%). The cooking rotation is therefore load-bearing: without it the two specialists would need 126 h/cycle. NEXUS absorbs the monitoring and scheduling that would otherwise push the non-galley line past its 49.0 h. Section 5 gives the accounting method and its assumptions.

Who goes out for the regolith, how many EVA hours, and how much mass per sol? And how is the +200 kcal per EVA hour provided?

Two numbers, because the requirement is not ours to scope. Rules p.12 requires the food system to provide an additional 837 kJ (200 kcal) per EVA hour for *crewmembers performing EVA operations* — all of them, not only the food-system tasks. ESPERANZA II therefore declares both what it drives and what it must absorb.

Attributable to the food system — 0.7 crew-EVA-h/sol. Regolith: HybridSoil is 21.3% Martian material, so 49.3 kg/sol of substrate requires **10.5 kg/sol** of Martian regolith, 5,250 kg over the rotation. (The 32.1% lunar fraction is inherited from the transit shielding mass and is not acquired by EVA — see Solution Summary §2.) Acquisition is not performed in three-minute increments: it is batched as 2 crew × 6 h every 30 sols, i.e. **0.40 crew-EVA-h/sol**. Radiator: 300 m² at an assumed 40 m² per suited crew-hour is 7.5 h per full clean, scheduled as 2 crew × 4 h every 30 sols, i.e. **0.27 crew-EVA-h/sol**; the cadence is anchored to the same Martian dust deposition that drives the α 0.15 → 0.55 derating in Section 1.

Mission envelope — 8.0 crew-EVA-h/sol nominal, design box to 24. Eight crew-EVA-hours per sol is one two-person eight-hour EVA every second sol: 4,000 crew-EVA-h over the rotation, and 5.6 EVA-h per week for each of ~10 EVA-qualified crew. The food-system share is 12× smaller than the envelope, which is the point — the constraint is set by the surface architecture, not by this system.

The supplement is produced in situ, and that is a compliance decision, not a preference. Appendix B Table 3 caps pre-packaged and bulk-ingredient calories at half of the total meal plan, and the EVA supplement is meal-plan energy: there is no exemption. Sourced from Earth, compliance would depend on a variable the food system does not control:

Crew-EVA-h/sol	Supplement (kcal/sol)	Earth-provisioned if supplied from Earth	Earth-provisioned if produced in situ
0 (baseline)	0	46.97%	46.97%
4	800	47.89%	46.16%
8 — nominal	1,600	48.77%	45.38%
12	2,400	49.63%	44.62%
13.8 — break point	2,757	50.00% — cap reached	44.29%
16	3,200	50.45% — non-compliant	43.89%
24 — design box	4,800	52.03% — non-compliant	42.49%

Read the last two columns against each other. Supplied from Earth, the 50% cap is breached at 13.8 crew-EVA-h/sol — a cadence set by the surface architecture, not by this food system. Produced in situ, the Earth-provisioned fraction **decreases monotonically** with EVA hours, reaching 42.49% at three times nominal. **Compliance is independent of the EVA profile by design.**

How it is provided, and where it is eaten. The requirement is to *provide* the energy, not to consume it inside the suit — and the suited state does not permit eating: the EMU carries only an in-suit drink bag of water, and at 8.2 psi / 34% O₂ the prebreathe window is minimal by design. For an 8-hour EVA the 1,600 kcal are delivered as a loading meal 60 min before egress (600 kcal, enlarged portions of the standard in-situ rotation), one 90 g compact bar at the airlock (300 kcal), water only in-suit per EMU heritage, and a recovery meal within the 30-minute glycogen resynthesis window (700 kcal, enlarged portions plus 25–30 g protein). **Roughly 80% of the supplement is therefore larger servings of food the system already produces:** no new process line, no new hardware, and the portioning falls to the all-crew rotation and the EVA team rather than to the two food specialists.

Two safety constraints are stated rather than discovered. Crumbs in a 34% O₂ atmosphere are both foreign-object debris and a flammability path: the bar is specified soft, bound and film-coated, and is consumed in the galley or airlock, never in the donning bay. And *Tenebrio* flour is an allergen (tropomyosin, cross-reactive with crustacean and mite): an insect-free variant of the bar is drawn from the allergen-free reserve for any affected crew member.

What it costs. 1,600 kcal/sol is +6.63% on the 24,141 kcal/sol in-situ line: +25 m² of illuminated canopy (373 → 398), +2.0 kW of LED, +13 m² of radiator, +0.6 L/sol of net water make-up, and the specialists move from 81.2 to 82.8 h in their worst 5-sol window — still inside 90 h, still no overtime. **Launch mass: zero.** This is declared as a sensitivity analysis on the baseline architecture, not as a re-optimised design. Two second-order couplings are stated for completeness: carbon fixation rises to 21.6 kg/sol, moving the Martian CO₂ make-up from 3.1 to 4.0 kg/sol; and on EVA sols roughly 0.33 kg/sol of crew-exhaled CO₂ is vented through the suit scrubber instead of returning to the canopy, which the same Martian make-up covers. With the 1,600 kcal/sol EVA supplement produced in situ, total demand is 45,525 + 1,600 = 47,125 kcal/sol, so the Earth-provisioned share is 21,384 / 47,125 = 45.4%. The 47.0% figure remains the baseline, reported against the 45,525 kcal/sol nameplate.

Where does the initial water charge come from?

It is inherited with the habitat. The 12,000 L nutrient-solution inventory is not mined and not shipped for this system: the ESPERANZA II crew is a relief crew taking over infrastructure already in operation, consistent with the mission scenario stated in the ConOps Phase 2. The food system does not generate water, it conserves it — 644 L/sol of canopy transpiration recovered at 98.5%, with 9.7 L/sol of make-up drawn from ECLSS grey water. Initial water procurement sits outside the food-system boundary, as does the power source.

What is the biometric loop actually based on? A per-astronaut gut-health inference is a strong claim at this TRL.

It is based on the author's own analysis of public NASA flight data, released openly and citable. Working the Artemis II Human Research Data Methodology Challenge on the SpaceX Inspiration4 archive (NASA OSDR, OSD-569 to OSD-687; n = 4 civilians, six timepoints L-92 to R+82), a label-permutation test showed that a fused multi-modal representation separates pre- from post-flight physiological state at LOSO accuracy 0.708 against a null of 0.489 ± 0.107 over 2,000 permutations — p = 0.026 (Puerta Angulo, 2026, DOI 10.21203/rs.3.rs-10232328/v1).

Two findings from that work shape NEXUS directly. First, a feature-efficiency frontier: 18 cardiovascular readouts were **not distinguishable** from 406 urine cytokine features (accuracy 0.679 vs 0.688, overlapping 95% credible intervals at n = 4 — non-distinguishability under low power, not demonstrated equivalence). Note the boundary of the claim: the significant p = 0.026 belongs to the **fused** representation, which includes the laboratory modalities; the cardiovascular modality alone was not shown significant. The wristband is therefore adopted as the minimum-crew-time sensor, not as a validated stand-alone predictor. NEXUS is therefore specified around a non-invasive wristband rather than a laboratory panel, and salivary cortisol is demoted from routine periodic sampling to confirmation only when the fused loop raises WATCH. Sampling is crew time, and crew time is the one constrained resource in Appendix B. Order of magnitude, stated as an estimate rather than a measurement: routine periodic sampling of 15 crew at one saliva sample per sol, at ~2.5 min per sample for collection, labelling and log entry, is ~0.63 h/sol. Event-driven confirmation at an assumed WATCH rate of ~2 events per five-sol week is ~0.02 h/sol. The net ~0.6 h/sol returns to the **49.0 h/cycle non-galley budget** of Section 5 — it does not alter the measured 15.40 h/sol galley load, which is derived from Deliverable 2.2 and is unchanged. The 2.5 min/sample and the WATCH rate are engineering assumptions to be confirmed in an analogue campaign, not measured values.

Second, an explicit limit on what may be claimed: no individual biomarker survived multiple-testing correction (0 of 1112 features at Benjamini–Hochberg q < 0.05), and calibration of the separate cardiovascular-modality hierarchical model was

honest rather than over-confident (Brier 0.224, mean p_{correct} 0.555 across four LOSO folds — deliberately near-chance confidence, which is the correct behaviour at $n = 4$, and not a property of the fused model above). NEXUS is accordingly specified as a fused *state* detector, not as a single-marker screen — in the author's words, "*at $n = 4$, the return on a single-marker screen is negligible.*" Stated limitations: the work is a preprint under review, and Inspiration4 is a short-duration low-Earth-orbit civilian analogue. It substantiates the inference method, not a transfer of the numerical result to Mars.

Does NEXUS require the crew to change their behaviour?

No. NEXUS proposes; the crew decides. Portion and macro personalisation is opt-out, and the nutritional targets of the 14-sol plan are met in full if the crew ignores the system entirely. It is an operations aid, not a dietary regime.

Why is 3.0 $\mu\text{mol/J}$ a defensible fixture efficacy?

It is achievable in current top-tier commercial horticultural fixtures at the wall plug, though at the favourable end of the range. At 2.7 $\mu\text{mol/J}$ canopy demand rises to approximately 33.8 kW mean, which the installed capacity of 77 kW accommodates without redesign.

Reconciliation statement. The figures in this appendix are cross-checked against one another, not only against their own sources. The 644 L/sol of transpiration is derived from the same 30.4 kW of lighting that drives the 58.1 kW power budget and the 300 m² radiator; the 41.2% Earth-provisioned fraction is derived from the same ingredient list that produces the 15.40 h/sol crew-time load. They are one physical system, and moving any one of them moves the others.